Mobility, Economic Conditions, and Crime During COVID-19: Evidence from Boston

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Abstract

This paper examines how reported crime in Boston changed during the COVID-19 pandemic, with a focus on whether mobility restrictions or economic distress better explain the short-run decline. Using monthly Boston Police Department data from 2018 to 2022, merged with FRED labor market data and Google Mobility reports, I estimate OLS regressions and tree-based machine learning models. The main finding is that reported crime fell sharply at the onset of lockdowns even as unemployment tripled, consistent with routine activity theory rather than an economic distress channel. Transit mobility is positively associated with crime in the stronger specifications, and the random forest importance matrix confirms that the COVID dummy alone has near-zero predictive power. The results suggest that disruptions to urban movement, not labor market stress, drove the short-run decline in crime.

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1. Introduction

Research on crime has long examined how economic conditions shape offending, with many studies focusing on unemployment, labor market stress, and opportunity cost. Classic economic models emphasize that weaker labor market opportunities may reduce the opportunity cost of offending and increase incentives for crime (Becker, 1968). Raphael and Winter-Ebmer (2001) find evidence that unemployment increases crime, while Gould, Weinberg, and Mustard (2002) show that weaker labor market opportunities are associated with higher criminal activity. This paper started from that framework, asking not just whether reported crime in Boston rose or fell during COVID-19, but how its composition changed over time, as both economic conditions and daily movement patterns were disrupted simultaneously.

That shift in focus matters because COVID-19 was not only a labor market shock. It also changed how people moved through the city, how often they used public transit, and how often potential offenders, victims, and guardians came into contact. Research in environmental criminology and routine activity theory emphasizes that crime opportunities depend on the convergence of these three elements in space and time (Cohen and Felson, 1979; Felson and Boba, 2010). Newton's overview of the urban mobility literature highlights that crime patterns are connected to the dynamic movement of people through urban space, making mobility an important explanatory factor rather than just background context.

This perspective is especially useful for studying crime during the pandemic. Mohler et al. argue that social distancing could change both the volume and distribution of crime by increasing guardianship in some settings while reducing public interaction in others. Hodgkinson and Andresen (2020) find that the pandemic affected crime types unevenly, consistent with shifts in activity patterns and opportunity structures. A broader systematic review by Hoeboer et al.

concludes that many crime categories fell during COVID restrictions, though effects differed across offense types and often reversed once restrictions eased. Ashby (2020) documents early evidence of crime declines during lockdowns in several U.S. cities, while Abrams (2021) provides one of the first empirical analyses linking COVID restrictions to crime changes.

More recent evidence strengthens this framing. Frith, Bowers, and Johnson show that changes in household occupancy during COVID restrictions help explain burglary patterns, while Trajtenberg et al. find across 45 cities that stricter lockdowns were associated with larger declines in robbery, burglary, and vehicle theft. Campedelli, Aziani, and Favarin (2021) similarly emphasize that mobility reductions and routine activity disruptions were central mechanisms. Together, these studies suggest that pandemic-era crime should be understood through both economic stress and changing patterns of mobility and routine activity.

This paper applies that framework to Boston using monthly Boston Police Department incident data from 2018 to 2022. It examines how reported crime and its composition changed before, during, and after the main COVID disruption. The analysis uses monthly crime counts together with unemployment, labor force, a COVID indicator, a time trend, and a Google mobility measure based on transit station activity. The goal is descriptive rather than strictly causal: to test whether changes in local economic conditions and mobility are systematically associated with changes in reported crime over time. The rest of the paper is organized as follows: Section 2 describes the data, Section 3 presents descriptive statistics and visualizations, Section 4 reports the regression results, Section 5 discusses the web-scraped COVID activity data, Section 6 presents the machine learning results, and Section 7 concludes.

2. Data

2.1. Boston Crime Data

The crime data were collected from separate Boston Police Department incident report files covering 2018 to 2022 and merged into a single dataset after standardizing column names and cleaning date fields. Invalid observations were removed and the incident date was used to aggregate data to the monthly level. Seven crime categories were defined based on offense descriptions: violence and assault, property-related, vehicle-related, investigation, medical assistance, drug-related, and other.

2.2. Labor Market Data

Labor market data were obtained from the Federal Reserve Economic Data (FRED) system and include monthly Boston-area unemployment rate and labor force size, restricted to the 2018 to 2022 sample window. The unemployment rate captures labor market stress, while the labor force variable captures broader changes in participation not fully reflected in the unemployment rate alone.

2.3. Mobility Data

Mobility data come from the Google Community Mobility Reports. The global file was filtered to Suffolk County, Massachusetts, and the transit stations percent change from baseline variable was extracted and aggregated to monthly averages. A key limitation is that this series is only available from February to December 2020, yielding 11 monthly observations after aggregation. Results from models including mobility should therefore be interpreted as suggestive rather than conclusive. After merging the three sources, a COVID-period indicator was created equal to one for all months from March 2020 onward, and a linear time trend was added. The final dataset is a monthly Boston panel with 60 observations covering January 2018 through December 2022.

3. Descriptive Statistics and Visualizations

3.1. Summary Statistics

Table 1 reports summary statistics for the main variables. Monthly reported crime averages about 6,709 incidents, with values ranging from 4,387 to 9,072, reflecting meaningful variation over the sample period. The unemployment rate varies widely, from 2.6 to 16.5 percent, capturing the major labor market disruption associated with COVID-19. Transit mobility, available only for 2020, shows an average decline of about 57 percent from baseline.

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Monthly reported crime	60	6,708.98	1,098.57	4,387	9,072
Unemployment rate (%)	60	4.62	3.10	2.60	16.5
COVID period dummy	60	0.57	0.50	0.00	1.0
Time trend	60	30.50	17.46	1.00	60.0
Transit mobility change (%)	11	-56.81	21.49	-80.03	0.4
Labor force	60	2,803,432	46,987	2,595,759	2,883,061

Note: Transit mobility data are available for 11 months in 2020 only.

Table 2 breaks the sample into three periods. Average reported crime falls from about 7,627 incidents per month pre-COVID to about 5,866 during the shock period, even as the unemployment rate rises sharply from 2.92 to 10.23 percent. This pattern is the central puzzle the paper addresses: crime fell at exactly the time economic conditions worsened most severely. Reported crime remains below its pre-COVID level in the post-COVID period, suggesting only a partial recovery.

Table 2. Means by Period

Period	Total Crime	Unemployment (%)	Labor Force	Transit Mobility (%)
Pre-COVID	7,627.35	2.92	2,839,844	0.4
COVID Shock	5,865.50	10.23	2,770,492	-62.53
Post-COVID	6,065.54	4.12	2,777,711	N/A

Note: Transit mobility is only available during the COVID shock period.

3.2. Key Figures

Figure 1 plots monthly reported crime in Boston from 2018 to 2022. Crime is relatively high before 2020, drops sharply at the onset of COVID-19, and then recovers only partially. The decline in early 2020 is much larger than any normal seasonal movement, which is why COVID is treated as a distinct structural break in the analysis.

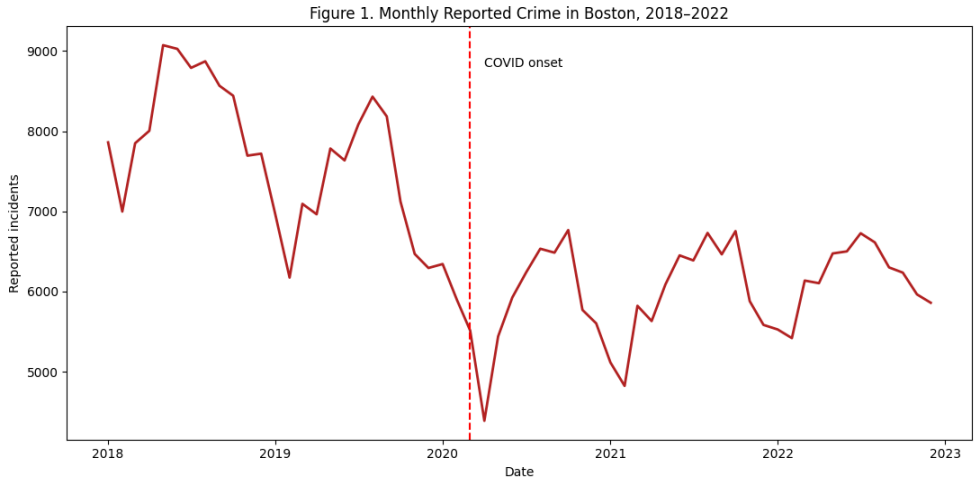

Figure 1. Monthly Reported Crime in Boston, 2018-2022

3.3. Main Message

The main message figure plots three series together during the 2020 mobility window (the only period for which Google Mobility data are available): crime as a percent change from February 2020, transit mobility as a percent change from baseline, and the unemployment rate. Around March 2020, reported crime drops sharply at the same time that transit mobility collapses and unemployment rises. The central tension is that crime declines even as economic distress

worsens, suggesting that mobility restrictions may have mattered more than unemployment in the short run.

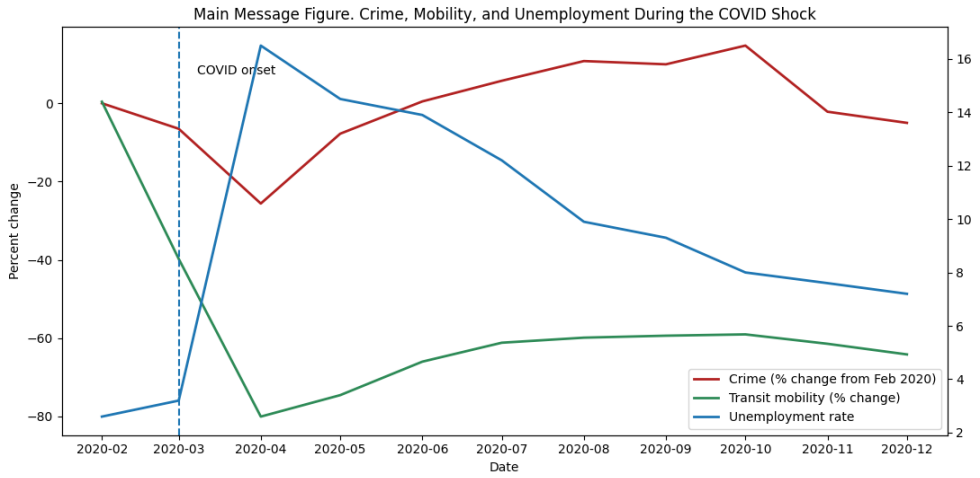

Figure 2. Crime, Mobility, and Unemployment During the COVID Shock

3.4. Crime Composition

Figure 3 shows how the composition of reported crime changed across categories over time. Property crime rises as a share of total incidents around March 2020, while vehicle-related crime drops sharply. This suggests that the lockdown did not affect all crime types equally: reduced commuting, lower traffic volume, and changes in daily routines altered the opportunity structure for different offenses.

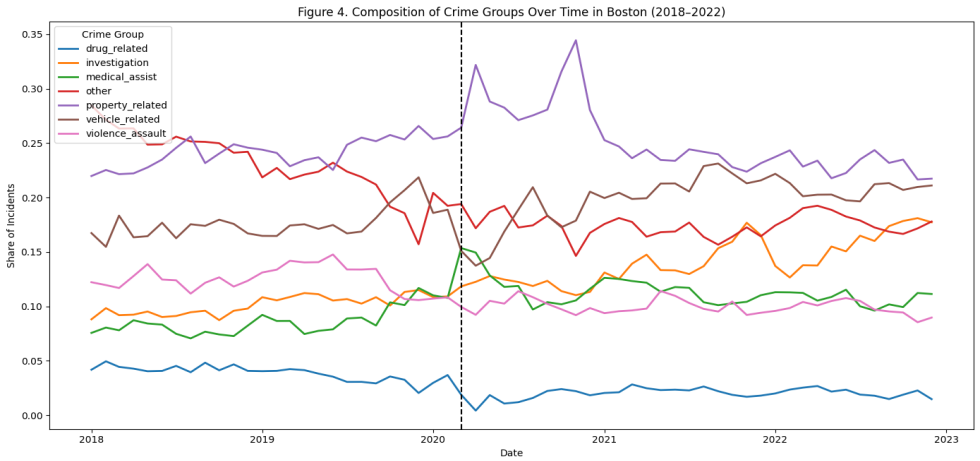

Figure 3. Composition of Crime Categories Over Time, 2018-2022

4. Regression Results

4.1. Mechanisms and Specifications

Two mechanisms could explain why crime fell during COVID despite rising unemployment. The first is the routine activity channel: with transit mobility collapsed by roughly 80 percent, the convergence of offenders, targets, and guardians in public space was dramatically reduced, cutting crime opportunities independent of economic conditions. The second is the economic distress channel: rising unemployment should, under classic theory, increase crime by lowering the opportunity cost of offending. The regression results consistently support the first channel and contradict the second.

The baseline OLS specifications take the following form, where t indexes months, covid is a dummy equal to one from March 2020 onward, and trend is a linear time trend:

(1) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \varepsilon_t$

(2) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot labor_force_t + \varepsilon_t$

(3) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \beta_4 \cdot labor_force_t + \varepsilon_t$

(4) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \beta_4 \cdot unemployment_rate_t^{\,2} + \varepsilon_t$

The mobility specifications are estimated on the 11-observation subsample where Google Mobility data are available:

(5) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot mobility_t + \varepsilon_t$

(6) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \beta_4 \cdot mobility_t + \varepsilon_t$

(7) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot labor_force_t + \beta_4 \cdot mobility_t + \varepsilon_t$

(8) $total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \beta_4 \cdot labor_force_t + \beta_5 \cdot mobility_t + \varepsilon_t$

4.2. Baseline Results

Table 3 below reports the baseline regression results using a full 60 month sample.

Table 3. Baseline OLS Specifications

	Dependent variable: total crime			
	(1)	(2)	(3)	(4)
Unemployment rate squared				-7.472
				(10.972)
Constant	8401.891***	-3613.575	-5375.280	7974.354***
	(331.718)	(8713.483)	(8373.514)	(710.793)
COVID period	-310.977	-1283.284***	28.691	-613.701
	(561.141)	(266.637)	(589.998)	(717.998)
Labor force		0.004	0.005	
		(0.003)	(0.003)	
Time trend	-31.887**		-34.624**	-26.263
	(14.150)		(14.037)	(16.442)
Unemployment rate (%)	-117.864**	-27.115	-99.892**	24.551
	(46.671)	(38.547)	(47.252)	(214.315)
Observations	60	60	60	60
R^2	0.594	0.570	0.613	0.597
Adjusted R^2	0.572	0.547	0.585	0.568
Residual Std. Error	718.716 (df=56)	739.419 (df=56)	707.980 (df=55)	722.182 (df=55)
F Statistic	27.282*** (df=3; 56)	24.745*** (df=3; 56)	21.765*** (df=4; 55)	20.381*** (df=4; 55)
Note:	*p<0.1; **p<0.05; ***p<0.01			

The clearest result is the negative time trend. In Models 1 and 3, the trend coefficient is approximately -31.9 and -34.6, both statistically significant, suggesting that reported crime was already declining by roughly 32 to 35 incidents per month on average. The unemployment rate is also negative and significant in Models 1 and 3, with coefficients of about -117.9 and -99.9. Higher unemployment is associated with lower reported crime, which contradicts a simple economic distress story and supports the routine activity channel: lower activity levels and reduced movement may have outweighed any short-run pressure from labor market stress.

The COVID coefficient is less stable across the baseline models. In Model 2, where COVID is included with unemployment and labor force but without a time trend, the coefficient is large, negative, and statistically significant at about -1,283. Once the time trend is added, the COVID coefficient becomes much smaller and loses significance, suggesting that part of the COVID effect is captured by the broader downward trend in the data.

4.3. Preferred Specification

My preferred specification is Model 3, which includes unemployment rate, the COVID dummy, a time trend, and labor force:

$$total_crime_t = \alpha + \beta_1 \cdot unemployment_rate_t + \beta_2 \cdot covid_t + \beta_3 \cdot trend_t + \beta_4 \cdot labor_force_t + \varepsilon_t$$

This specification has the highest adjusted R-squared among the baseline models at 0.585 across 60 observations, and includes all theoretically motivated variables without introducing the redundancy of the squared unemployment term in Model 4.

4.4. Mobility Results

Table 4 reports the regression results including the transit mobility variable.

Table 4. Mobility and Expanded OLS Specifications

	Dependent variable: total crime			
	(1)	(2)	(3)	(4)
Constant	5817.340***	-1243.746	-22853.192*	-4801.054
	(775.876)	(1797.195)	(9399.557)	(14419.445)
COVID period	2187.323	4437.828**	-722.424	3708.593
	(2279.014)	(1390.209)	(1810.639)	(3296.581)
Labor force			0.010**	0.002
			(0.003)	(0.006)
Transit mobility change (%)	38.257	128.192***	-25.884	107.113
	(46.776)	(34.314)	(37.944)	(92.530)
Time trend		243.031***		212.872
		(60.027)		(137.624)
Unemployment rate (%)	24.753	296.410**	-20.128	255.758
	(123.311)	(96.206)	(84.603)	(193.967)
Observations	11	11	11	11
R^2	0.215	0.790	0.693	0.792
Adjusted R^2	-0.122	0.649	0.488	0.584
Residual Std. Error	698.745 (df=7)	390.683 (df=6)	472.126 (df=6)	425.342 (df=5)
F Statistic	0.637 (df=3; 7)	5.627** (df=4; 6)	3.380* (df=4; 6)	3.810* (df=5; 5)
Note:	*p<0.1; **p<0.05; ***p<0.01			

The mobility models add the transit mobility variable. In Model 6, transit mobility has a positive and statistically significant coefficient of about 128.2, suggesting that months with higher transit activity are associated with higher reported crime. This supports the routine activity interpretation: more movement in public space means more crime opportunity. The mobility results are not stable across all four specifications, which is expected given the short sample, but the direction is consistent throughout.

5. Web Scraping and COVID Activity

5.1. Data Source and Approach

To supplement the main datasets, COVID-related data were scraped from the Wikipedia page on the COVID-19 pandemic in Massachusetts. The table reporting dated COVID-related entries was selected for its clear time dimension. The date field was extracted and transformed into a monthly count of entries, serving as a measure of scraped COVID activity over time. This variable is not used in the main regressions but provides supporting descriptive evidence.

5.2. Visualization

Figure 4 compares scraped COVID activity to monthly reported crime in Boston. The sharpest crime changes coincided with the most intense phase of COVID-related reporting activity in early 2020, supporting the idea that the main crime decline was concentrated in the initial lockdown period.

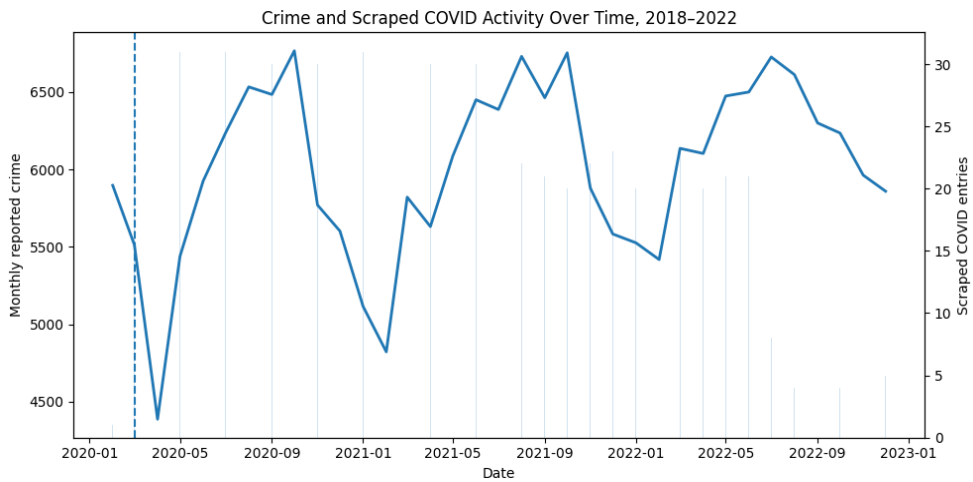

Figure 4. Crime and Scraped COVID Activity Over Time, 2018-2022

6. Machine Learning Results

6.1. Regression Tree

To explore nonlinear relationships between crime and its predictors, a regression tree was estimated using unemployment rate, labor force, the COVID dummy, and the time trend as features, with a maximum depth of three. The objective function minimized by the tree is:

$$min\, \Sigma_m\, \Sigma_{i+rm}\, (y_i - \bar{y}R_m)^2$$

where y_i is the monthly crime count and $\bar{y}R_m$ is the mean within each leaf region. The tree identifies threshold effects in the data, splitting first on the time trend and then on labor force and unemployment rate, revealing that crime responds differently depending on economic conditions rather than following a single global linear relationship. It is important to note that this analysis is descriptive and is used to detect nonlinear patterns in the data, not to make causal claims about the effect of any individual variable on crime.

6.2. Random Forest and Variable Importance

The random forest importance matrix provides additional insight. Figure 5 shows that the time trend is the dominant predictor, followed by labor force and unemployment rate. Most notably, the COVID dummy alone has near-zero importance. This suggests that the pandemic affected crime primarily through underlying time trends and activity-related changes, rather than as a standalone binary shock. This finding reinforces the regression results: what drove crime changes during COVID was the collapse of mobility and routine activity, not the pandemic label itself.

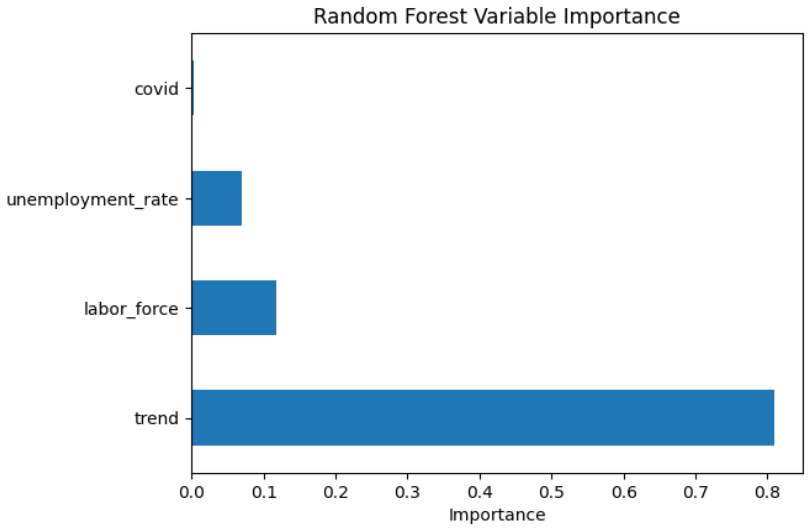

Figure 5. Random Forest Variable Importance

6.3. OLS vs. Machine Learning

Compared to OLS, the tree-based models add two things. First, they reveal nonlinear threshold effects that a linear model cannot capture without explicit interaction terms. Second, the importance matrix provides a model-free ranking of which variables matter most, confirming and extending the regression findings rather than contradicting them.

7. Conclusion

This paper examined how reported crime in Boston changed during the COVID-19 period, with particular attention to whether the pandemic altered not only the level of crime but also its composition. The main finding is that reported crime fell sharply during the initial lockdown period, even as unemployment rose. This pattern is difficult to explain using a simple economic distress framework alone. The evidence in this paper suggests that the pandemic changed crime partly by disrupting routine activity and reducing movement through the city.

The main message figure captures this clearly. Around March 2020, crime drops at the same time that transit mobility collapses and unemployment increases. These three series together suggest that the short-run effect of lockdowns and reduced movement may have been stronger than the upward pressure from worsening labor market conditions. Fewer people moving through public space likely meant fewer opportunities for some types of crime, even during a period of substantial economic stress.

The spatial evidence also showed that the decline in crime was uneven across districts, which is consistent with mobility and routine activity changing differently across the city rather than with a uniform economic shock affecting all neighborhoods equally. The regression results point in the same broad direction. In the baseline models, crime trends downward over time and higher unemployment is not associated with higher reported crime. In the mobility models, transit mobility is positively associated with crime in one of the stronger specifications, supporting the idea that movement through the city mattered for short-run crime outcomes. The machine learning results add further texture: the random forest identifies the time trend, labor force, and unemployment rate as the most important predictors in the preferred specification, while the COVID dummy alone has near-zero importance.

There are important limitations to acknowledge. The mobility analysis is based on only 11 observations, since the Google Mobility dataset only covers part of 2020, limiting the robustness of those regressions. In future work, a longer mobility series would allow a more convincing test of the mechanism. It would also be valuable to build offense-specific models, separating property crime, violence, and vehicle-related incidents, to understand which crime types were most sensitive to mobility restrictions versus economic stress. Collecting data on policy stringency by date could further help identify the timing and magnitude of the COVID shock.

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